

Sarthak Bichhawa

Sr. Data & Applied Scientist · Microsoft, Azure Storage — Staff / Principal Applied Scientist

Redmond, WA · sarthaksgsits@gmail.com · linkedin.com/in/sarthak-bichhawa · github.com/sarthak0501 · sarthak0501.github.io

Sr. Data & Applied Scientist with 9+ years of production ML/AI. Ships autonomous multi-agent LLM systems (planner/executor on Azure OpenAI + MCP, Sev2 triage hours to minutes), a fleet-scale behavioral embedding platform (autoencoder + FAISS, millions of accounts), and the eval harnesses and data-quality guardrails that let both be trusted in production.

EXPERIENCE

Sr. Data & Applied Scientist MICROSOFT · REDMOND, WA

Nov 2020 – present

AI & AGENTIC SYSTEMS

- Cut customer-health time-to-insight from days to <2 minutes with an enterprise LLM agent for Mission-Critical Customers — natural-language questions become governed live Kusto/ADX queries with PM-ready narratives, grounded on a curated catalog of 13 metric domains.
- Reduced multi-hour on-call incident investigations to minutes with an autonomous multi-agent investigation agent — planner/executor LLM loop on Azure OpenAI with context compaction, confidence scoring, and run tracing (FastAPI on Azure).
- Shipped two reusable Model Context Protocol (MCP) tool servers — one almost entirely AI-generated — serving governed query and telemetry tools to Copilot, VS Code, and internal agents.

APPLIED ML & DATA-QUALITY PLATFORM

- Enabled fleet-scale similarity search, segmentation, and embedding-drift anomaly detection across millions of accounts by designing an account-behavior embedding model (autoencoder + FAISS) that compresses each Azure Storage account into a vector fingerprint.
- Caught silent upstream data regressions — including an ~88% row-collapse — before they reached dashboards or Finance, with a self-service data-quality monitoring service (Python/FastAPI, anomaly detection, 76 tests, Azure Container Apps) adopted across teams, plus hourly table-health profiling over 47 datasets.

RELIABILITY & EXPERIMENTATION

- Built an adaptive pre-release stress-testing program for Azure Storage informing per-build ship/hold decisions; helped prevent 10+ Sev1/Sev2 customer incidents by catching top-severity regressions before rollout.
- Ran thousands of controlled simulations per build across 20 workload types, distilling ~500 failure signals into 15 recurring failure modes and confirming 50+ issues with reproduction steps and quantified impact.
- Prioritized high-yield test scenarios with online learning (multi-armed bandits) and tuned scoring thresholds on outcomes, cutting false positives 60%; led a 2-person effort and authored the design doc + internal white paper.

PARTNER REVENUE & BUSINESS REPORTING

- Recovered and sustained ~\$45M/month in influenced partner revenue (<1% variance vs Finance, ~4x partner coverage) by rebuilding a 20+ component attribution pipeline with multi-partner credit replacing last-writer-wins — lifting partner-attributed influenced revenue >10% month-over-month.
- Collapsed monthly business-review reporting from a multi-day manual process to a governed ~5-minute pipeline (>95% effort reduction), now in regular leadership use.

MISSION-CRITICAL CUSTOMER HEALTH

- Drove the MCC health program and a unified customer-health view across 8+ engineering/support teams, cutting executive/customer readiness insight from 3–5 days to near real-time — with ~50% fewer >10-day aging issues and ~16% faster time-to-mitigate.
- Helped reverse churn risk for 10+ critical customers — each representing multi-millions in monthly Azure Storage revenue — by accelerating early detection and action on customer-impacting issues.
- Shipped 15+ Azure Advisor recommendations adopted across hundreds of thousands of customer accounts to improve cost and performance.

LEADERSHIP

- org-recognized AI Champion — drove AI adoption across the team
- co-led a 90-minute internal AI Day-of-Learning
- sustained code-review mentoring + onboarded a new hire
- led 4-team cross-org Sev2 incident response
- 8+ engineering/support teams aligned on one health program

Senior Data Scientist WALMART LABS · BENTONVILLE, AR

Feb 2019 – Nov 2020

- Delivered ~7% category revenue lift across 50 stores with an end-to-end planogram optimization system, cutting planning cycles from weeks to hours.
- Built product association + substitution analytics (behavior tree + demand-transfer "universes") to support assortment and replenishment decisions.
- Led 4 DS/analysts + 2 interns to productionize a repeatable item-level placement workflow; earned a Spot Award.

Data Scientist Intern INTEGRAL AD SCIENCE · NEW YORK, NY

Jun – Aug 2018

- Built a PySpark contextual text-classification pipeline processing ~1M URLs/day across 846 categories at 88% accuracy; improved label quality via correlation analysis.

Data Scientist MU SIGMA · BENGALURU, INDIA

Oct 2015 – Jul 2017

- Estimated customer lifetime value with probabilistic and regression models; improved competitive reporting by unifying product IDs via fuzzy matching.

SKILLS

AI / ML Generative & agentic AI · multi-agent orchestration (planner/executor, tool-calling, RAG/grounding) · MCP servers & clients · Azure OpenAI · LLM observability, tracing & evals · vector embeddings & autoencoders · FAISS · anomaly detection · multi-armed bandits · NLP classification · production ML

TOOLS Python · SQL · Kusto (ADX) · PySpark · R · FastAPI · Power BI · Tableau · Azure DevOps · Git

PLATFORMS Azure (App Service, Container Apps, Data Factory) · large-scale telemetry & data-quality pipelines
· experimentation & causal measurement · revenue attribution & business reporting

EDUCATION

UNIVERSITY OF ILLINOIS AT CHICAGO MS, Business Analytics · 2017–2018

SGSITS, INDORE B.E., Computer Engineering · 2011–2015